

Curcumin-HP-beta-CD Study

Abstract

The solubility of curcumin increased from 0.6 microgram per mL to 3.2 mg per mL upon complexation with HP-beta-CD at a 1:2 molar ratio suggesting a 1:1 inclusion complex.

Results

The phase solubility showed AL-type with stability constant 1250 per molar. Figure 1 shows FT-IR spectra. This

suggests
the
aromatic
ring
is
inserted
into
the
hydrophobic
cavity
of
HP-beta-CD.
Table
1
summarizes
solubility
enhancement.
However
in
vitro
results
may
not
directly
predict
in
vivo
performance
and
further
studies
are
warranted.

Conclusion

HP-beta-CD
significantly
enhanced
aqueous
solubility.
Future
work
should
investigate
in
vivo
bioavailability.